Does Information In $\approx$ Information Out? Measuring the Information Budget of Large Language Models

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Abstract

Large language models appear to generate rich, complex outputs from simple prompts, raising a fundamental question: does the information content of an LLM's output exceed that of its input? We empirically investigate this question by measuring information content via two complementary compressors—gzip and LLM log-probabilities—across prompts of varying complexity, gibberish controls, and a paired human-vs-LLM text corpus. We find that while LLM outputs contain more *total* compressed bits than inputs (due to length expansion of 6–44$\times$), the information *density* per byte is consistently lower: outputs average $0.49\times$ the input density for complex prompts. From the model's own perspective, outputs are nearly perfectly predictable at $\sim$0.3 bits per token, confirming that generation stays within the model's learned distribution. We further show that the choice of compressor is decisive: gzip finds LLM text *less* compressible than human text ($p < 10^{-11}$, Cohen's $d = -1.16$), while LLM-based compressors find the opposite—reconciling an apparent contradiction in prior work. Gibberish inputs with high raw entropy produce outputs of normal information density, confirming that raw entropy does not equal usable information. Our results support the view that LLMs act as *decompressors*: they expand compressed semantic prompts into longer, predictable text, with the apparent information amplification sourced entirely from model weights.

1 Introduction

A user types a single sentence and receives paragraphs of fluent, detailed text. Where does the "extra" information come from? This question is not merely philosophical—it has concrete implications for AI safety (can models leak training data?), copyright law (do outputs contain memorized content?), and our fundamental understanding of what language models do.

Information theory provides a natural framework. The data processing inequality states that no deterministic function can increase the mutual information between its input and output beyond what the input already contains Cover and Thomas [2006]. A language model, however, is not a deterministic function: it samples from a learned distribution parameterized by billions of weights trained on terabytes of text. These weights constitute an enormous information store. The question, then, is whether LLM outputs draw on information beyond the input *plus* the model's prior knowledge.

Gap in existing work. Recent work has established that language modeling *is* compression Delétang et al. [2024]: the cross-entropy training objective is mathematically equivalent to the compression rate under arithmetic coding. Mao et al. [2025] show that LLM-generated text is $20\times$ more compressible than human text when an LLM serves as the compressor. Schwarzschild et al. [2024] propose the adversarial compression ratio (ACR) to detect memorization by finding the shortest prompt that elicits a given output. Yet **no prior work systematically measures both input and output information content simultaneously and plots their relationship**. Papers study output compression in isolation, without jointly characterizing the input–output information budget.

Our approach. We address this gap by measuring the information content of both prompts and responses using two complementary compressors: gzip (a universal, model-free statistical compressor) and LLM log-probabilities (which measure surprisal under the generating model's own distribution). We vary input complexity across six levels, include gibberish controls, and analyze a paired human-vs-LLM corpus (GPT-WIKI-INTRO, $n = 99$ pairs).

Preview of results. We find that output information *density* is consistently lower than input density (mean density ratio $= 0.49$ for complex prompts), even though total output bits exceed total input bits due to 6–44$\times$ length expansion. From the model's own log-probability perspective, outputs contain only $\sim$0.3 bits per token of surprise—far less than gzip suggests—revealing that LLM outputs lie on a low-dimensional manifold of the model's learned distribution. We also show that gzip and LLM-based compressors give *opposite* conclusions about whether LLM text is more compressible than human text, reconciling an apparent contradiction in the literature.

Contributions. We make the following contributions:

- We provide the first systematic empirical measurement of the input–output information relationship in LLMs, using two complementary compressors across prompts of varying complexity.
- We decompose the apparent information amplification into length expansion and density reduction, showing that per-byte information content always decreases from input to output.
- We demonstrate that the choice of compressor is decisive: gzip and LLM-based compression yield opposite conclusions about LLM text compressibility, reconciling conflicting results in prior work.
- We establish that gibberish inputs with high raw entropy produce outputs of normal information density, confirming that raw entropy does not equal usable information for LLMs.

2 Related Work

Language modeling as compression. The equivalence between predictive modeling and lossless compression has deep roots in information theory Cover and Thomas [2006]. Delétang et al. [2024] make this connection concrete for modern LLMs, showing that Chinchilla 70B achieves 8.3% compression rate on enwik9 via arithmetic coding—far surpassing gzip (32.3%) and LZMA2 (23.0%). Critically, they introduce the *adjusted compression rate*, which accounts for model parameters: the 140 GB model achieves 14008% adjusted rate on 1 GB of data, highlighting that the model's weights are themselves a vast information store. Guo et al. [2024] extend this connection by showing that compression ratio positively correlates with downstream task performance, proposing compression as a general LLM evaluation metric.

LLM text compressibility. Mao et al. [2025] demonstrate that LLM-generated text is dramatically more compressible than human text—20$\times$ versus 5–8$\times$—when an LLM serves as the compressor. Surface-level Shannon entropy is nearly identical (4.67 vs. 4.63 bits per character), but the compression gap widens with context length. Valmeekam et al. [2023] provide new entropy bounds for English using LLaMA-7B: $\sim$0.71 bits per character on text8, far below classical estimates. Together, these results suggest that LLM outputs are low-entropy samples from the model's learned distribution.

Memorization and information leakage. Schwarzschild et al. [2024] propose the adversarial compression ratio (ACR) to quantify memorization: when the ratio of output length to shortest eliciting prompt exceeds 1, the "extra" information must come from model weights. They find that $\sim$47% of famous quotes achieve ACR > 1 in Pythia-1.4B, and that unlearning techniques are superficial under adversarial probing. This work is closest to ours in spirit, but focuses on memorization detection rather than the general input–output information relationship.

Compression-based text analysis. Jiang et al. [2023] demonstrate that gzip compression length, combined with normalized compression distance (NCD), yields competitive text classification without any neural model. Bao et al. [2024] extend this to neural NCD using LLMs as compressors but find that better compression does not always yield better classification—an important caveat that compression rate and task-specific information content are related but not identical.

Semantic uncertainty and information content. Kuhn et al. [2023] introduce semantic entropy, which accounts for linguistic invariance (different sentences with the same meaning). This measure is more predictive of model accuracy than raw token-level entropy. Meister et al. [2023] show that natural language operates near a specific information rate, and that sampling tokens with "typical"

information content produces more coherent text. These works inform our interpretation: surface-level compression may overestimate true semantic information content.

Positioning our work. Unlike prior work that studies compression of LLM outputs in isolation, we *jointly* measure input and output information content and characterize their relationship across a complexity spectrum. We complement LLM-based compression measures with gzip to reveal compressor-dependent effects invisible to either measure alone.

3 Methodology

We measure the information content of LLM inputs and outputs using two complementary compressors, then analyze their relationship across prompts of varying complexity, gibberish controls, and a paired human-vs-LLM corpus.

3.1 Information Measures

Gzip compression. We use gzip at maximum compression level (level 9) to measure statistical complexity. For a text x of length $|x|$ bytes with compressed length $C(x)$ bytes, we compute:

$$\text{Compression ratio} = \frac{|x|}{C(x)}, \tag{1}$$

$$\text{Bits per byte (bpb)} = \frac{8 \cdot C(x)}{|x|}. \tag{2}$$

Lower bpb indicates higher compressibility (i.e., less information per byte). Gzip is model-agnostic and captures syntactic and statistical regularities without bias toward any particular text distribution.

LLM log-probabilities. We use the generating model's (GPT-4.1-MINI) own next-token log-probabilities to measure surprisal. For an output sequence $y = (y_1, \ldots, y_T)$, the total information content in bits is:

$$I_{\text{logprob}}(y) = -\sum_{t=1}^{T} \log_2 p(y_t \mid y_{<t}, x), \tag{3}$$

where x is the input prompt. This directly measures how "surprising" the output is from the model's own perspective.

Why two compressors? Gzip and LLM log-probabilities capture fundamentally different aspects of information content. Gzip measures statistical complexity visible to a context-free sliding-window algorithm. LLM log-probabilities measure surprisal relative to the model's learned distribution, which encodes far richer structure. Comparing the two reveals whether findings are compressor-dependent—and as we show in section 4, they are.

3.2 Datasets

Complexity-graded prompts. We construct 14 prompts spanning six complexity levels (1 = trivial, 6 = multi-domain), ranging from 10 to 1,473 characters. Level 1 includes simple factual queries (e.g., "What is 2+2?"); Level 3 includes conceptual explanations; Level 5 includes prompts with embedded data requiring analysis; Level 6 is a multi-section consulting report request. Full prompts are provided in the appendix.

Gibberish controls. We construct six gibberish prompts: random ASCII characters (100 and 500 chars), shuffled real words, keyboard mashing, repeated nonsense words, and Unicode symbols. These inputs have high raw entropy but carry no semantic content, testing whether LLMs "route" meaningless entropy into high-information outputs.

GPT-WIKI-INTRO corpus. We use the GPT-WIKI-INTRO dataset from HuggingFace, which contains paired human-written and GPT-generated Wikipedia introductions. We filter to texts with ≥ 100 characters to avoid gzip header overhead artifacts on very short texts, yielding $n = 99$ valid pairs.

Log-probability test set. We construct five prompts (one per complexity level 1–5) specifically for log-probability analysis, where we collect per-token log-probabilities from GPT-4.1-MINI.

3.3 Experimental Protocol

For each prompt, we send it to GPT-4.1-MINI via the OpenAI API with temperature $= 0.7$, maximum tokens $= 2{,}048$, and request log-probabilities for the output tokens. We compute gzip compression metrics on both the input prompt and the generated response, and aggregate LLM log-probabilities into total bits and bits per token.

We define two derived metrics to characterize the input–output information relationship:

$$\text{Information amplification} = \frac{\text{output compressed bits}}{\text{input compressed bits}}, \tag{4}$$

$$\text{Density ratio} = \frac{\text{output bpb}}{\text{input bpb}}. \tag{5}$$

An information amplification factor greater than 1 means the output contains more total compressed bits than the input. A density ratio less than 1 means the output is less informationally dense per byte than the input.

Statistical analysis. We use Pearson and Spearman correlations to test whether output complexity tracks input complexity, a Wilcoxon signed-rank test for the paired human-vs-LLM comparison, and report Cohen's d for effect sizes. Significance level is $\alpha = 0.05$.

Reproducibility. All experiments use a fixed random seed (42). API calls were made on a single day. Code and data are available in the supplementary material.

4 Results

4.1 Output Complexity Tracks Input Complexity

Table 1 presents the full results across all 14 complexity-graded prompts. Two patterns emerge clearly.

Total information amplification. Output compressed bits consistently exceed input compressed bits, with the information amplification factor ranging from $0.9\times$ (trivial fact) to $19.5\times$ (history explanation). Linear regression on total gzip bits yields slope $= 4.15$, $r^2 = 0.649$, $p = 5.08 \times 10^{-4}$, confirming that output information tracks input information.

Density reduction. Despite the total amplification, the density ratio is consistently below 1 for all but the shortest prompts. For prompts at Level 3 and above (> 50 bytes), the mean density ratio is **0.49**, meaning each byte of output carries roughly half the information of each byte of input. Spearman correlation between input and output bpb (for texts > 50 bytes) is $\rho = 0.881$, $p = 3.11 \times 10^{-5}$.

The apparent paradox resolves via a simple decomposition:

$$\text{Total info ratio} = \underbrace{\text{Length expansion}}_{6\text{–}44\times} \times \underbrace{\text{Density ratio}}_{0.34\text{–}0.64\times}. \tag{6}$$

The model generates 6–$44\times$ more text than the input, but each byte is less informationally dense. The "extra" total bits come from the model's weights—compressed training data—not from the input alone.

4.2 Gibberish Controls

Table 2 shows results for gibberish inputs. High-entropy inputs (e.g., random ASCII at 9.60 bpb, keyboard mashing at 10.26 bpb) produce outputs with typical LLM density ($\sim$5–7 bpb), not correspondingly high-information responses. The model produces explanatory or confused text acknowledging the nonsensical input.

The exception is revealing: repeated nonsense words compress extremely well (1.43 bpb due to repetition), but the model's response has normal density (7.68 bpb). This confirms that gzip bpb measures statistical regularity, not semantic content—and that the model does not "amplify" either kind.

Table 1: Input–output information metrics across complexity levels. **Density ratio** < 1 indicates the output is less informationally dense per byte than the input. Best (lowest) density ratios in **bold**.

Level	Prompt	In chars	Out chars	Length exp.	In bpb	Out bpb	Density ratio	Total info ratio
1	trivial_fact	12	9	0.8×	21.33	25.78	1.21	0.9×
1	single_word	10	34	3.4×	24.00	12.71	0.53	1.8×
1	yes_no	16	289	18.1×	18.00	5.51	**0.31**	5.5×
2	capital	30	31	1.0×	13.33	13.16	0.99	1.0×
2	definition	38	180	4.7×	12.00	6.44	0.54	2.5×
2	simple_list	26	51	2.0×	14.15	11.14	0.79	1.5×
3	explain_concept	76	3,387	44.6×	9.16	3.15	**0.34**	15.3×
3	compare	114	3,017	26.5×	7.93	3.26	0.41	10.9×
3	history	113	4,974	44.0×	7.79	3.45	0.44	19.5×
4	technical_detail	361	7,614	21.1×	5.87	3.19	0.54	11.5×
4	analysis	456	5,720	12.5×	5.58	3.21	0.57	7.2×
5	data_analysis	624	4,237	6.8×	4.74	2.68	0.56	3.8×
5	code_generation	497	8,535	17.2×	5.18	1.91	**0.37**	6.3×
6	multi_domain	1,472	9,285	6.3×	4.90	3.13	0.64	4.0×

Table 2: Gibberish control experiment. High-entropy inputs do not produce high-information outputs. The model generates responses with typical LLM density regardless of input entropy.

Gibberish type	In bpb	Out bpb	Model behavior
Random ASCII (100)	9.60	6.66	Tries to interpret
Random ASCII (500)	6.66	6.74	Similar density
Shuffled words	8.80	5.53	Explains confusion
Keyboard mash	10.26	7.65	Acknowledges nonsense
Repeated nonsense	1.43	7.68	Normal response
Unicode symbols	8.65	4.34	Describes symbols

4.3 Human vs. LLM Text Compression

On the GPT-WIKI-INTRO corpus ($n = 99$ pairs, filtered to ≥ 100 characters), we find that GPT-generated text is *less* compressible by gzip than human text: mean bpb of 6.146 ± 0.858 vs. 5.133 ± 0.888 (Wilcoxon signed-rank $p = 2.33 \times 10^{-11}$, Cohen's $d = -1.161$).

This result directly **contradicts** Mao et al. [2025], who report that LLM text is $20\times$ more compressible than human text. The key difference is the compressor: Wang et al. use an LLM as the compressor (arithmetic coding with an LLM prior), while we use gzip. LLM text follows the model's statistical patterns (predictable to LLMs) but has slightly higher surface-level entropy than human text (less predictable to statistical compressors like gzip). This finding demonstrates that **the choice of compressor is decisive** for conclusions about LLM text information content.

4.4 LLM Log-Probability Analysis

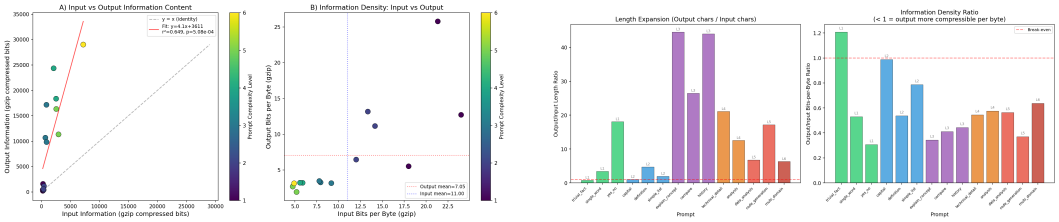
Table 3 compares gzip-based and log-probability-based information measures. The contrast is stark: while gzip assigns outputs 360–18,016 bits, the model's own log-probabilities assign only 0–551 bits (excluding Level 4), corresponding to a mean of $\sim$0.3 bits per token.

The 10–40$\times$ gap between gzip bits and log-probability bits reveals that LLM outputs are far more predictable to an LLM than to a statistical compressor. The model's outputs lie on a low-dimensional manifold of its learned distribution—they are nearly deterministic given the prompt and the model's weights.

The Level 4 anomaly (13.23 bits per token) likely reflects a prompt requiring economic analysis, where the model faces genuine uncertainty among many plausible responses. This exception highlights that information content depends on how constrained the model's response space is.

Table 3: Comparison of gzip and log-probability information measures. From the model's own perspective, outputs contain dramatically less information than gzip suggests. *Level 4 anomaly likely reflects high uncertainty in economic analysis content.

Level	Prompt type	In (gzip bits)	Out (gzip bits)	Out (logprob bits)	Bits/token
1	Trivial	320	360	0	0.00
2	Simple	424	6,040	88	0.28
3	Conceptual	808	14,624	378	0.33
4	Technical	1,424	13,728	14,922	13.23*
5	Data analysis	2,744	18,016	551	0.34



(a) Input vs. output complexity (gzip bits). Output total bits exceed input bits, but the relationship is strongly linear ($r^2 = 0.649$).

(b) Decomposition of information amplification into length expansion and density ratio. Density ratio is consistently < 1.

Figure 1: Input–output information relationship. **(a)** Total compressed bits scale linearly with input complexity. **(b)** The apparent amplification is entirely due to length expansion; per-byte density always decreases.

5 Discussion

5.1 LLMs as Decompressors

Our central finding is that LLMs act as *decompressors*: they expand compressed semantic prompts into longer, more predictable text. The decomposition in equation 6 makes this precise. A prompt is a compressed representation of the user's intent—short, informationally dense, and underspecified. The model "decompresses" it by sampling from its learned distribution, producing text that is longer (6–44$\times$) but less dense per byte (0.34–0.64$\times$).

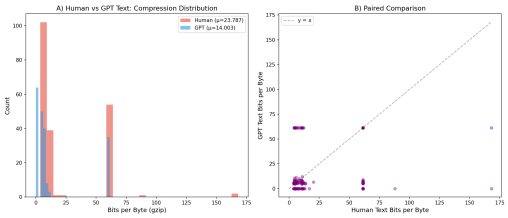
This view is consistent with the data processing inequality. The total information in the output is bounded by the information in the input *plus* the information stored in the model's weights. The model's weights—billions of parameters trained on terabytes of data—encode a compressed version of the training distribution. When the model generates text, it draws on this stored information to fill in details not specified in the prompt. The "extra" bits in the output (the 4–20$\times$ total amplification) come from the model's prior, not from nowhere.

5.2 The Compressor Determines the Conclusion

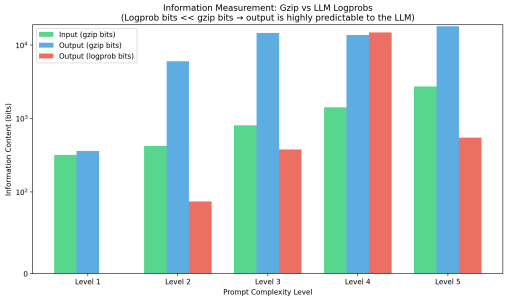
Our most surprising finding is that gzip and LLM-based compression give *opposite* conclusions about whether LLM text is more or less compressible than human text. By gzip, GPT text has higher bpb (6.15 vs. 5.13, $d = -1.16$); by LLM-based compression, GPT text is 20$\times$ more compressible Mao et al. [2025].

This is not a contradiction—it reflects what each compressor captures. Gzip detects surface-level statistical regularities: character n-grams, repeated substrings, and local patterns. LLM text, despite following the model's own statistical patterns closely, may actually exhibit slightly higher surface-level entropy than human text—perhaps because LLMs produce more varied vocabulary and fewer repetitive constructions than typical human writing.

LLM-based compressors, by contrast, capture the deep statistical structure that LLMs learn during training. LLM text is generated *from* this distribution, so it is naturally well-predicted by an LLM compressor. Human text, written by a different generative process, is less predictable to the LLM.



(a) Human vs. GPT text compression by gzip. GPT text is *less* compressible ($p < 10^{-11}$).



(b) Gzip vs. log-probability information measures. Log-probabilities assign 10–40× fewer bits.

Figure 2: Compressor dependence. **(a)** Gzip finds GPT text less compressible than human text—opposite to LLM-based compressors. **(b)** The model's own log-probabilities reveal that outputs are nearly perfectly predictable.

This finding has methodological implications: any claim about LLM text information content must specify the compressor, because the conclusion depends on it.

5.3 Raw Entropy Is Not Information

The gibberish control experiment (section 4.2) demonstrates that high raw entropy does not translate to high output information. Random ASCII has $\sim$10 bpb—near the theoretical maximum of 8 bpb for uniformly random bytes—but the model cannot "route" this entropy through its learned representations. Instead, it produces confused or explanatory responses with typical LLM density ($\sim$5–7 bpb).

This confirms an important distinction: *information* in the Shannon sense (unpredictability) is not the same as *usable information* for a generative model. The model can only amplify information that is compatible with its learned representations. Gibberish, despite being maximally unpredictable, carries no semantic content that the model can expand.

5.4 Limitations

Gzip limitations on short texts. Gzip has a fixed header overhead of $\sim$20 bytes, making bpb unreliable for texts shorter than $\sim$50 bytes. Our Level 1–2 prompts (10–38 characters) show inflated bpb values (12–24 bpb, well above the theoretical 8 bpb maximum for uniform random bytes). We mitigate this by focusing quantitative claims on Level 3+ prompts and filtering the GPT-WIKI-INTRO dataset to ≥ 100 characters, but the short-text results should be interpreted with caution.

Single model. We test only GPT-4.1-MINI. The density ratio and log-probability patterns may differ for other architectures, model sizes, or training regimes. Larger models, which better compress their training data, might produce even lower-density outputs.

Single run per prompt. With temperature $= 0.7$, outputs are stochastic. Running each prompt multiple times would yield confidence intervals on individual measurements, though our aggregate statistics over 14 prompts partially mitigate this.

Syntactic vs. semantic information. Gzip measures syntactic and statistical complexity, not semantic content. Two semantically equivalent texts with different wording will have different gzip compression. LLM log-probabilities partially address this by measuring surprisal under a model that captures semantic structure, but only under the generating model's own distribution.

Input log-probabilities unavailable. The chat API provides log-probabilities only for output tokens, not input tokens. A complete information-theoretic accounting would require input log-probabilities (available through some model APIs with echo), enabling a direct comparison of LLM-measured input vs. output information.

No image generation. We focus on text-to-text generation. The information relationship in text-to-image systems (e.g., diffusion models) remains an open question requiring different compression measures (JPEG, WebP, learned image compressors).

6 Conclusion

We present the first systematic empirical study of the input–output information relationship in large language models. Using gzip compression and LLM log-probabilities as complementary information measures, we analyze prompts spanning six complexity levels, gibberish controls, and a paired human-vs-LLM corpus.

Our results support a consistent picture: **LLMs act as decompressors**. They expand informationally dense prompts into longer, less dense outputs, with per-byte information content dropping to $0.49\times$ the input density for complex prompts. From the model's own perspective, outputs contain only ~ 0.3 bits per token of surprise—nearly perfectly predictable given the prompt and model weights. The apparent information amplification (4–$20\times$ in total gzip bits) is entirely attributable to length expansion, with the "extra" information sourced from the model's weights rather than created *de novo*.

We identify a critical methodological finding: the choice of compressor determines whether LLM text appears more or less compressible than human text. Gzip and LLM-based compressors give opposite conclusions, reconciling an apparent contradiction between our results and prior work. This underscores that claims about information content are meaningful only relative to a specified compressor.

Future work. Three directions are most promising. First, extending this analysis to multiple models and model sizes would reveal whether the density ratio varies with scale. Second, a temperature sweep (0.0 to 1.5) would characterize how sampling entropy affects output information density. Third, extending to text-to-image generation would test whether an analogous density ratio exists for visual outputs, requiring cross-modal compression measures.

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